

2024 - 2025 Junior Science & Humanities Symposium Statement of Outside Assistance



Students submitting their research paper to the Regional and National symposium must complete this form in full and submit with the final research paper.

Please type "N/A" in any field that is "not applicable" to your research.

Projects conducted without proper supervision will be disqualified from both regional and National competition. Further guidelines may be found at <http://www.jshs.org> & in the Core Rules of Competition.

Student Participant to Complete:

Name: Vincent Lu

Title of Paper: Exploring Innovative Machine Learning Models for Soil CO2 Emission Forecasting

School: Basis Independent Fremont

Teacher or Mentor Name: NA

Regional Symposium: California Northern

1. Please explain your role in the development of the project idea.

I developed the project idea myself.

2. What steps led you to formulate your research question? – or – What steps led you to develop the design for your project?

There have been lots of studies in forecasting soil respiration. However, most of them are process-based models that are computationally intensive.

Can machine learning models, such as XGBoost and Deep Neural Network models, be used to predict soil respiration accurately?

Many factors affect soil respiration. What are the key factors in driving soil respiration and how?

Can I provide explainability for those machine learning models that are usually considered as black box models?



Junior Science and Humanities Symposium
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- 3. Where did you conduct the major part of your work? If an institutional setting, list the name.**
(e.g., home, school, or other institutional setting such as university lab, medical center, etc.)

Home

- 4. Describe the assistance that you received throughout the project.**

I did the research myself. No assistance from others.

- 5. If you conducted your research in an institutional setting (i.e., university lab, medical center, etc.), describe your role on the team AND what role each person played in the research investigation?**

N/A



6. **Describe what parts of the research you did on your own and what parts where you received help.** (e.g., literature search, hypothesis, experimental design, use of special equipment, gathering data, evaluation of data, statistical analysis, conclusions, and preparation of written report (abstract and/or paper))
I did all parts by myself.

7. **Was any data set from an outside source (literature, handbooks, publicly available or privately shared tables, etc.) used in the research? If so, what data set(s) was used and from where was it sourced? What was its purpose in this research?**

Yes. I used the Soil Respiration Database Version 5 (SRDB-V5), a comprehensive open-source database of published research data on soil respiration. I used the dataset to build machine learning models to predict CO₂ emission form soil respiration.

8. **Was the data set sourced outside this research the only way to obtain the data required for the research? Please explain.**

No. Process-based models and experiments can be used to study soil respiration. However, those approaches are time-consuming and computationally intensive. Leveraging open-source databases and machine learning models provides an effective and powerful way to forecast CO₂ emissions from soil respiration based on observable environmental factors.

- 9. Is this research a continuation of an investigation that was previously submitted to a regional JSHS? If so, describe how you have expanded your investigation.**

No

- 10. If this is a continuation project, please submit your previously submitted abstract here (copy/paste).**

N/A



11. Did you utilize any form of AI in your research or paper? If so, explain what tools were used AND for what purpose.
I did not use AI in my research.

12. Is there an IRB determination on file for this research? (Yes/No)
No

13. Provide the IRB determination details including IRB number, name, institution, and dates.
N/A

14. Provide full details about the processes and procedures utilized for your research, including usage & disposal of materials.

Phase 1: Project preparation

- Literature review
- Identify research gaps
- Propose solution
- Search for database

Phase 2: ML model development

- Data preparation
- Feature selection
- Model architecture
- Model training
- Model evaluation

Phase 3: Model explainability and Deep dive

- Model performance comparison
- SHAP analysis for model explainability

Materials used: Python, Sklearn, XGBRegressor, SRDB dataset, Tensorflow, SHAP, Random Forest Regressor

Teacher and/or Supervising Mentor to Complete (or Parent if no teacher/mentor involvement):

Comments by teacher and/or supervising mentor on the students' individual contributions to the research investigation or engineering/computer science project. If no Teacher or Mentor/Scientist was involved the Parent must complete this section describing their role in the research.

Vincent Lu did the research by himself. No help was provided by parents. -- Yi He

Statement by the teacher and/or supervising mentor acknowledging that the student conducted the research in accordance with proper procedures and protocols for the conduct of animal research or human research.

N/A

The Student, Teacher and/or Supervising Mentor must sign below. If you did the work without a teacher or supervising scientist, you will need a signature from your parent and a brief description of their role in the research.

1/15/2025

Date

Vincent Lu

Signature of Student (required)

Basis Independent Fremont

Student's High School

Date

Signature of Teacher

Teacher's High School

Date

Signature of Supervising Mentor

Name of Supervising Mentor

Title of Supervising Mentor

Institution of Supervising Mentor

1/15/2025

Date

Yi He

Signature of Parent (required if no Teacher or Supervising Mentor was involved in the research.)

Yi He (408-341-5573)

Name and Phone Number of Parent